

SCUBE3-Related Short Stature Syndrome (SSFSC2): Comprehensive Disease Characterization

Disease: SCUBE3-Related Short Stature Syndrome **Formal name:** Short stature, facial dysmorphism, and skeletal anomalies with or without cardiac anomalies 2 (SSFSC2) **MONDO:** MONDO:0030953 · **OMIM:** 619184 · **Category:** Mendelian (autosomal recessive) **Date compiled:** 2026-09-05

Summary

SCUBE3-Related Short Stature Syndrome (SSFSC2) is an **ultra-rare, autosomal-recessive Mendelian disorder** caused by **biallelic loss-of-function variants in *SCUBE3*** (Signal peptide-CUB-EGF domain-containing protein 3; HGNC:23310, NCBI Gene:222663, chromosome 6p21.31, UniProt Q8IX30). The disorder was first defined as a discrete clinical entity by Lin and colleagues in 2021, who described 18 affected individuals from 9 unrelated families sharing a consistent phenotype of reduced growth, skeletal features, a distinctive craniofacial appearance, and dental anomalies ([PMID: 33308444](#)). As of 2025, approximately 20 patients from around 10 families have been reported worldwide ([PMID: 40331102](#)).

Mechanistically, SCUBE3 is a secreted/cell-surface-anchored glycoprotein that functions as a **co-receptor for bone morphogenetic proteins BMP2/BMP4**, recruiting BMP receptor complexes into lipid-raft microdomains and augmenting BMP–BMP type I receptor interactions. It also modulates TGF- β (via SMAD2/3), FGF, and Hedgehog signaling. Biallelic loss-of-function impairs transcript processing, protein secretion, and co-receptor function, dysregulating BMP signaling. Because BMP-mediated chondrogenesis and osteogenesis are essential for endochondral bone formation at the growth plate, SCUBE3 deficiency produces prenatal-onset short stature and skeletal/craniofacial/dental abnormalities. The *Scube3*^{-/-} knockout mouse faithfully recapitulates the human disorder, exhibiting craniofacial and dental defects, reduced body size, and defective endochondral bone growth due to impaired BMP-mediated chondrogenesis and osteogenesis ([PMID: 33308444](#)).

Clinically, the syndrome is **non-lethal with generally normal cognition and normal life expectancy**. Diagnosis is molecular—recognition of the clinical gestalt followed by exome/genome sequencing or a short-stature/skeletal-dysplasia gene panel. There is **no disease-modifying therapy**; management is supportive and multidisciplinary (orthopedic/spinal, dental/orthodontic, cardiac surveillance, growth monitoring). The GH/IGF-1 axis is typically normal, so the growth failure reflects a local growth-plate BMP-signaling defect rather than classic growth-hormone deficiency.

Section 1 — Disease Information

Overview. SSFSC2 is a syndromic short-stature disorder of prenatal onset combining growth retardation, a recognizable craniofacial gestalt (long/triangular face, high forehead, long nose, small pointed chin), skeletal anomalies (brachydactyly, short long bones, scoliosis, 11 rib pairs), and dental anomalies (crowding, enamel defects), typically with preserved cognition ([PMID: 33308444](#)).

Key identifiers.

Resource	Identifier
MONDO	MONDO:0030953
OMIM	619184
DOID	DOID:0112358
UMLS	C5543057
MedGen	1782253
Causal gene	<i>SCUBE3</i> — HGNC:23310, NCBI Gene:222663, Ensembl ENSG00000146197, UniProt Q8IX30
Locus	6p21.31
Inheritance	Autosomal recessive (HP:0000007)

Synonyms / alternative names: Short stature, facial dysmorphism, and skeletal anomalies with or without cardiac anomalies 2 (SSFSC2); SCUBE3-related short stature syndrome; SCUBE3-related developmental disorder; SCUBE3 deficiency. Gene aliases: *SCUBE3*, *CEGF3*.

Data source type: Information is derived from **aggregated disease-level resources** (OMIM, MONDO, HPO annotations) built on **individual-patient case series** from the primary literature; there is no large EHR-derived dataset for this ultra-rare condition. No dedicated Orphanet ORPHA code or MeSH term has been identified; MONDO:0030953 / OMIM:619184 are the canonical entries.

Section 2 — Etiology

Primary cause: Purely genetic—**biallelic (homozygous or compound heterozygous) loss-of-function variants in SCUBE3**. There is **no environmental, infectious, or multifactorial component**; this is a monogenic germline disorder ([PMID: 33308444](#)).

"bi-allelic inactivating variants in SCUBE3 have pleiotropic consequences on development and cause a previously unrecognized syndromic disorder. Eighteen affected individuals from nine unrelated families showed a consistent phenotype characterized by reduced growth, skeletal features, distinctive craniofacial appearance, and dental anomalies." — [PMID: 33308444](#)

Genetic risk factors: The causal variants are inactivating *SCUBE3* alleles (nonsense, canonical splice-site, exon-skipping deletions, and functionally validated missense variants). **Consanguinity is a major driver**, with homozygous variants reported in Turkish, Iranian, Saudi/Emirati, Indian, Israeli, and Brazilian families. No modifier genes have been established.

Environmental risk factors / protective factors / gene-environment interactions: None known or applicable. No lifestyle, occupational, toxic, or dietary factors have been implicated, and no protective alleles or environmental protective factors are reported.

Section 3 — Phenotypes

Phenotype frequencies below are drawn from HPO disease annotations for OMIM:619184, sourced from the founding cohort (~15 evaluable individuals; [PMID: 33308444](#)). The phenotype is prenatal-onset, non-progressive in its core growth deficit, and of variable expressivity.

"Eighteen affected individuals from nine unrelated families showed a consistent phenotype characterized by reduced growth, skeletal features, distinctive craniofacial appearance, and dental anomalies." — Lin et al. 2021

Phenotype	HPO term	Frequency	Type
Postnatal growth retardation / short stature	HP:0008897	~100% (15/15)	Physical/growth
High forehead	HP:0000348	~100% (15/15)	Craniofacial sign
Long face	HP:0000276	~100% (14/14)	Craniofacial sign
Triangular face	HP:0000325	~100% (14/14)	Craniofacial sign
Short chin	HP:0000331	~100% (14/14)	Craniofacial sign
Long nose	HP:0003189	~87% (13/15)	Craniofacial sign
Prominent nasal bridge	HP:0000426	~87% (13/15)	Craniofacial sign
Pointed chin	HP:0000307	~85% (11/13)	Craniofacial sign
Dental crowding	HP:0000678	80% (12/15)	Dental anomaly
Short palm	HP:0004279	80% (12/15)	Skeletal
Brachydactyly	HP:0001156	80% (12/15)	Skeletal
Microcephaly	HP:0000252	~69% (9/13)	Craniofacial
Intrauterine growth retardation	HP:0001511	~73% (8/11)	Growth, antenatal
Antenatal onset	HP:0030674	~73% (8/11)	Temporal
Thick vermilion border	HP:0012471	~47% (7/15)	Craniofacial
Short long bone	HP:0003026	~55% (6/11)	Skeletal
Scoliosis	HP:0002650	~46% (6/13)	Skeletal
Enamel hypoplasia	HP:0006297	~27% (4/15)	Dental
11 pairs of ribs	HP:0000878	~31% (4/13)	Skeletal
Oligodontia / hypodontia	HP:0000677 / HP:0000668	~13% (2/15 each)	Dental
Brittle hair	HP:0002299	~13% (2/15)	Ectodermal
Pierre-Robin sequence	HP:0000201	~14% (2/14)	Craniofacial

Phenotype	HPO term	Frequency	Type
Atrial septal defect	HP:0001631	~14% (2/14)	Cardiac (variable)
Global developmental delay	HP:0001263	~7% (1/15)	Neurodevelopmental
Intellectual disability	HP:0001249	~7% (1/15)	Neurodevelopmental

Key clinical points: - **Cognition is usually normal** (developmental delay/intellectual disability each ~7%), an important distinguishing feature. - **Cardiac anomalies are variable** (ASD, arrhythmia, patent foramen ovale)—hence "with or without cardiac anomalies" in the disease name. - Occasional/rare features: joint hypermobility or stiffness, nystagmus/strabismus/astigmatism, hypotonia, cleft palate, hypospadias, tracheobronchomalacia, bicoronal craniosynostosis, oropharyngeal dysphagia. The expanded ocular/skeletal spectrum (high-arched eyebrows, epicanthus, blepharoptosis, hypotelorism, radial bowing, irregular vertebral endplates, narrow iliac wings) was added by [PMID: 40331102](#).

Severity/progression: Short stature is congenital and stable rather than progressive; scoliosis may progress and requires surveillance. **Quality-of-life impact** derives chiefly from short stature, skeletal (spinal) issues, and dental problems; disease-specific QoL instrument data (EQ-5D/SF-36/PROMIS) are not available.

Section 4 — Genetic / Molecular Information

Causal gene: *SCUBE3* (HGNC:23310, NCBI Gene:222663, 6p21.31, transcript NM_152753.4).

Pathogenic variant spectrum. ClinVar (accessed 2026) lists ~13 Pathogenic + 1 Pathogenic/Likely-pathogenic + 1 Likely-pathogenic *SCUBE3* records among ~184 total (63 VUS, 4 benign/LB). Disease-associated variants converge on **loss of function**:

Variant (NM_152753.4)	Protein	Class
c.1557C>A	p.Cys519Ter	Nonsense
c.2785C>T	p.Arg929Ter	Nonsense
c.1717C>T	p.Arg573Ter	Nonsense (conflicting)
c.2599+2T>C	—	Canonical splice
c.2239+1G>A	—	Canonical splice
c.829+3_952+2del	—	Exon-skipping deletion
c.2444T>C	p.Ile815Thr	Missense (Path/LP)
c.611G>A	p.Gly204Asp	Missense
c.291C>G	p.Cys97Trp	Missense (cysteine)
c.908G>C	p.Cys303Ser	Missense (disrupts EGF disulfide)
NG_050574.1 delins-inversion	—	Structural

Contiguous-gene **6p21 CNV deletions** (e.g., chr6:34.4–38.4 Mb) have also been reported (with severe IUGR). All variants are **germline**.

Functional consequences. In vitro validation showed variants variably impair transcript processing, protein secretion, and function, dysregulating BMP signaling:

"In vitro functional validation studies demonstrated a variable impact of disease-causing variants on transcript processing, protein secretion and function, and their dysregulating effect on bone morphogenetic protein (BMP) signaling." — PMID: 33308444

Population constraint (gnomAD, ENSG00000146197): pLI = 1.0; LOEUF (oe_lof_upper) = 0.378; oe_lof = 0.283 (33 observed vs 117 expected LoF); missense Z = 3.72. The gene is strongly constrained yet tolerant of heterozygous LoF alleles at low frequency—consistent with rare recessive carriers. No founder mutation established.

Modifier genes / epigenetics: No modifier genes or disease-specific epigenetic mechanisms have been reported. **Chromosomal abnormalities:** the contiguous-gene 6p21 microdeletions noted above.

Section 5 — Environmental Information

Not applicable. SSFSC2 is a **monogenic germline disorder with no known environmental, lifestyle, or infectious contributors**. No toxins, radiation, occupational exposures, dietary factors, or pathogens are implicated.

Section 6 — Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic loss-of-function SCUBE3 variants** (nonsense, splice, exon-skip, or destabilizing missense) → **reduced/absent functional SCUBE3 protein**, either through nonsense-mediated decay, impaired transcript processing, or defective secretion of misfolded protein. (*Demonstrated in vitro*; [PMID: 33308444](#).)
2. Destabilizing missense variants (e.g., p.Cys303Ser) → **disruption of a critical disulfide bridge in the 7th EGF-like repeat**, impairing protein stability and reducing secretion. (*AlphaFold3 modeling*; [PMID: 40331102](#).)
3. Loss of cell-surface SCUBE3 co-receptor → **failure to recruit BMP receptor complexes into lipid-raft microdomains** and reduced augmentation of BMP–BMP type I receptor interactions → **decreased BMP2/BMP4 signaling**. (*Demonstrated*; [PMID: 33308444](#).)
4. Reduced BMP signaling (and altered TGF- β /SMAD2-3 co-modulation) → **impaired SMAD-mediated transcriptional output** in growth-plate cells. (*BMP branch demonstrated*; *TGF- β branch inferred from CUB-domain ligand activity*, [PMID: 21441952](#).)
5. **Impaired chondrogenesis and osteogenesis** at the endochondral growth plate → **defective endochondral bone growth**. (*Demonstrated in Scube3^{-/-} mouse*; [PMID: 33308444](#).)
6. **Branch A:** growth-plate defect → **prenatal-onset short stature, short long bones, brachydactyly, scoliosis, rib anomalies**. **Branch B:** cranial neural-crest/dental mesenchyme BMP defect → **distinctive craniofacial gestalt and dental anomalies**. **Branch C (variable):** cardiac BMP signaling perturbation → **septal defects / arrhythmia** in a minority.

Detail

Molecular pathways. The core lesion is in the **BMP2/BMP4 signaling pathway** (SMAD1/5/8 branch), with SCUBE3 acting as a positive co-receptor:

"We show that SCUBE3 acts as a BMP2/BMP4 co-receptor, recruits the BMP receptor complexes into raft microdomains, and positively modulates signaling possibly by augmenting the specific interactions between BMPs and BMP type I receptors." — PMID: 33308444

SCUBE3 additionally acts as an **endogenous TGF- β type II receptor ligand** through its C-terminal CUB domain, inducing SMAD2/3 phosphorylation:

"bound to transforming growth factor- β (TGF- β) type II receptor through the C-terminal CUB domain, activated TGF- β signaling and triggered the epithelial-mesenchymal transition (EMT). This process includes the induction of Smad2/3 phosphorylation" — PMID: 21441952

Recombinant SCUBE3 boosts BMP2/TGF- β expression and drives osteogenic differentiation via SMAD phosphorylation in human bone-marrow mesenchymal stem cells, reinforcing its pro-osteogenic role (PMID: 39250278). SCUBE family members also participate in **FGF** (zebrafish muscle development) and **Hedgehog** signaling.

Protein dysfunction. SCUBE3 (~993 aa) has five modular domains:

"organized into five modular domains: (1) an N-terminal signal peptide sequence, (2) nine tandem epidermal growth factor (EGF)-like repeats, (3) a large spacer region, (4) three cysteine-rich (CR) motifs, and (5) a CUB domain at the C-terminus" — PMID: 37237303

The cysteine-rich, disulfide-stabilized EGF-like repeats are essential for folding and secretion; pathogenic missense variants that eliminate cysteines (Cys97Trp, Cys303Ser, Cys519Ter) destabilize the protein and reduce secretion. Structural modeling confirms this:

"Protein modeling using AlphaFold3 revealed disruption of a critical disulfide bridge within the seventh epidermal growth factor-like repeat, likely affecting protein stability." — PMID: 40331102

Cellular processes and cell types. Upstream: chondrocytes and osteoblasts of the endochondral growth plate; cranial neural-crest-derived mesenchyme; odontoblasts/dental mesenchyme. Downstream clinical manifestations follow from impaired proliferation/differentiation in these lineages. Suggested **GO** terms: BMP signaling pathway (GO:0030509), chondrocyte differentiation (GO:0002062), osteoblast differentiation (GO:0001649), endochondral ossification (GO:0001958), SMAD protein signal transduction (GO:0060395). Suggested **CL** terms: chondrocyte (CL:0000138), osteoblast (CL:0000062), odontoblast (CL:0000060), neural crest cell (CL:0000333).

Metabolic / immune / epigenetic involvement. No primary metabolic derangement, immune dysregulation, or epigenetic mechanism is part of the core disease; the single interesting metabolic observation is that recombinant SCUBE3 activates mitophagy and ameliorates oxidative stress during osteogenesis in vitro ([PMID: 39250278](#)), which is peripheral to the germline disease.

Section 7 — Anatomical Structures Affected

Organ / system level (primary): Skeletal system—long bones, vertebral column, ribs, hands (brachydactyly), skull/craniofacial skeleton. Endocrine-independent growth failure (GH/IGF-1 axis normal). **Secondary/variable:** cardiovascular system (atrial septal defect, PFO, arrhythmia); dentition; occasionally airway (tracheobronchomalacia), palate, and eyes.

Suggested UBERON terms: skeletal system (UBERON:0001434), endochondral bone (UBERON:0002513), epiphyseal growth plate (UBERON:0002332), long bone (UBERON:0002495), vertebral column (UBERON:0001130), rib (UBERON:0002228), skull (UBERON:0003129), tooth (UBERON:0001091), heart (UBERON:0000948), interatrial septum (UBERON:0002085).

Tissue / cell level: Connective tissue—cartilage (growth-plate chondrocytes) and bone (osteoblasts); dental mesenchyme/odontoblasts; cranial neural crest derivatives. **CL terms** as above.

Subcellular level: Secreted/cell-surface (plasma-membrane-associated) glycoprotein; biosynthetic defects implicate the **endoplasmic reticulum** (folding/secretion, GO:0005783) and **extracellular space / lipid-raft plasma-membrane microdomains** (GO:0045121) where SCUBE3 co-receptor function occurs.

Localization / lateralization: Skeletal and craniofacial involvement is **bilateral and largely symmetric**; scoliosis introduces axial asymmetry; cardiac defects are structural (septal).

Expression sites: primary osteoblasts, long bones, axial-skeleton cartilage, vascular endothelium, platelets, and mammary ductal epithelium ([PMID: 37237303](#), [PMID: 40331102](#)).

Section 8 — Temporal Development

- **Onset: Congenital / prenatal.** Intrauterine growth retardation and antenatal onset are documented in the majority (8/11 each). Short stature is present from birth.
 - **Onset pattern:** Chronic/insidious—a developmental, non-inflammatory process.
 - **Progression:** The core growth deficit is **stable (non-progressive)** rather than degenerative; the child remains short-statured. **Scoliosis may progress** through childhood/adolescence and requires monitoring.
 - **Disease course:** Lifelong, chronic, non-episodic, non-relapsing.
 - **Critical periods:** The prenatal and childhood growth-plate windows are the period of vulnerability; there is no known therapeutic window that reverses the growth-plate signaling defect.
 - **Remission:** None (genetic, structural condition).
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Section 9 — Inheritance and Population

- **Inheritance: Autosomal recessive** (HP:0000007), fully penetrant in reported biallelic cases with **variable expressivity** ([PMID: 33308444](#), [PMID: 40331102](#)). Most cases are homozygous; some are compound heterozygous.
- **Epidemiology: Ultra-rare.** ~20 patients from ~10 families reported worldwide as of 2025; no prevalence or incidence estimate exists.

"is a very rare genetic disorder caused by biallelic pathogenic variants in the SCUBE3 gene and has been reported in approximately 20 patients to date" — [PMID: 40331102](#)

- **Consanguinity:** A major driver—homozygous variants in Turkish, Iranian, Saudi/Emirati, Indian, Israeli, and Brazilian families.

- **Carrier frequency:** Rare; gnomAD shows only 33 heterozygous LoF alleles (LOEUF 0.38). No founder mutation.
- **Penetrance / anticipation / mosaicism:** Complete penetrance for biallelic genotypes; no genetic anticipation (not a repeat-expansion disorder); no germline mosaicism reported.
- **Sex ratio:** No sex bias; both sexes affected.
- **Population demographics:** No population-specific prevalence beyond enrichment in consanguineous populations; no established geographic clustering of specific variants.

Section 10 — Diagnostics

Diagnostic approach — molecular. Recognition of the **clinical gestalt** (prenatal-onset short stature + long/triangular face, high forehead, long nose, small pointed chin + brachydactyly + dental crowding/enamel defects with preserved cognition) followed by confirmation of **biallelic SCUBE3 variants**: - **Whole-exome sequencing (WES)** — the original discovery method; high yield ([PMID: 33308444](#)). - **Whole-genome sequencing (WGS)** — captures splice/structural variants. - **Multigene short-stature / skeletal-dysplasia panels** including *SCUBE3*; *SCUBE3* now appears on SGA/short-stature panels ([PMID: 41001785](#)). - **Single-gene testing** once the phenotype is recognized. - **Chromosomal microarray (CMA)** — detects contiguous-gene 6p21 deletions.

Supportive workup: skeletal survey/radiographs (brachydactyly, short long bones, scoliosis, 11 rib pairs, narrow iliac wings, vertebral endplate irregularities); echocardiography (ASD/PFO); dental/orthodontic evaluation; ophthalmologic exam; hearing assessment. **Endocrine (GH/IGF-1) axis is typically normal**—this is *not* classic GH deficiency, and there is **no specific biomarker**.

Differential diagnosis: 3M syndrome, Silver-Russell syndrome, Floating-Harbor syndrome, other primordial/skeletal-dysplasia short-stature disorders, ACAN- and COL2A1-related short stature. Preserved cognition, the specific facial gestalt, and biallelic *SCUBE3* findings distinguish SSFSC2.

Screening: Carrier/cascade testing in affected consanguineous families; prenatal/preimplantation genetic testing where the familial variants are known. No population newborn screening exists (ultra-rare).

Section 11 — Outcome / Prognosis

- **Survival / mortality: Non-lethal;** life expectancy is generally normal. No disease-specific mortality data reported.
 - **Morbidity / function:** Principal morbidity is **short stature, skeletal (scoliosis)** and **dental** problems; cognition is generally preserved, so functional/educational outcomes are usually good.
 - **Quality of life:** Impacted mainly by stature, orthopedic, and dental issues; no formal QoL instrument data available.
 - **Complications:** Scoliosis progression, dental crowding/enamel defects requiring orthodontic care, occasional cardiac defects requiring follow-up, rare airway issues (tracheobronchomalacia).
 - **Prognostic factors:** Severity is variable and not strongly predicted by a single factor; no validated prognostic biomarkers exist.
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Section 12 — Treatment

No disease-modifying or targeted therapy exists. Management is supportive, multidisciplinary, and symptom-directed:

Domain	Intervention	Suggested NCIT concept
Orthopedic	Spinal/scoliosis monitoring and management; orthopedic surgery as needed	Orthopedic Surgery
Dental	Orthodontic care for crowding, enamel defects, oligo/hypodontia	Dental Care
Cardiac	Echocardiographic surveillance; management of ASD/arrhythmia	Supportive Care Intervention
Growth	Growth monitoring; recombinant human GH has been <i>tried</i> with unproven/limited efficacy	Growth Hormone Therapy
Genetics	Genetic counseling for families	Genetic Counseling
Supportive	Developmental support if needed; ophthalmology/audiology follow-up	Supportive Care Intervention

Important caveat: Because the growth failure reflects a **local growth-plate BMP-signaling defect rather than GH deficiency**, recombinant human growth hormone is of unproven benefit. No pharmacogenomic, gene-therapy, cell-therapy, RNA-based, immunotherapy, or clinical-trial (NCT) interventions are currently available for this indication.

Section 13 — Prevention

Primary prevention of the disease itself is not possible (germline monogenic). Relevant preventive/genetic measures: - **Genetic counseling** for at-risk consanguineous families (25% recurrence risk per pregnancy for carrier couples). - **Carrier / cascade screening** of relatives once familial variants are identified. - **Prenatal diagnosis / preimplantation genetic testing** for known familial variants. - **Tertiary prevention** (preventing complications): scoliosis surveillance, dental prophylaxis/orthodontics, cardiac follow-up. No immunization, behavioral, environmental, or public-health interventions are applicable.

Section 14 — Other Species / Natural Disease

- **Taxonomy / orthologs:** Highly conserved across vertebrates ([PMID: 37237303](#)). Mouse *Scube3* (NCBIGene:268935, chr17), rat *Scube3* (NCBIGene:294297), zebrafish *scube3* (NCBIGene:100000687).
 - **Natural disease: No naturally occurring SCUBE3 disease reported in companion animals or wildlife** (no OMIA entry). No breed-specific (VBO) associations.
 - **Comparative biology:** Disease mechanism (BMP co-receptor function in skeletal development) is evolutionarily conserved, enabling faithful mouse and zebrafish modeling.
 - **Transmission:** Not applicable—no zoonotic or infectious component.
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Section 15 — Model Organisms

Model	Type	Key phenotype	Reference
<i>Scube3</i> ^{-/-} knockout mouse	Mammalian, null	Craniofacial & dental defects, reduced body size, defective endochondral bone growth from impaired BMP-mediated chondrogenesis/osteogenesis—recapitulates the human disorder	PMID: 33308444
<i>Scube3</i> ^{N294K} / <i>N294K</i> ENU missense mouse (German Mouse Clinic)	Mammalian, missense	Skeletal morphology abnormalities, altered bone-metabolism parameters, changed renal function, hearing impairment, plus energy-metabolism/behavioral/neurological alterations; resembles Paget disease of bone (6p21 is a PDB locus)	PMID: 27815347
Earlier <i>Scube3</i> -null mouse	Mammalian, null	Expressed in multiple developing tissues; dispensable for embryonic survival	PMID: 23383134
Zebrafish <i>scube3</i>	Teleost	Modulates FGF signaling during fast-muscle development; SCUBE family shows redundant/novel developmental roles	PMID: 24849601 , PMID: 40912446
Human BMSC in vitro	Cellular	rhSCUBE3 promotes osteogenic differentiation, mitophagy via BMP2/TGF- β -SMAD	PMID: 39250278

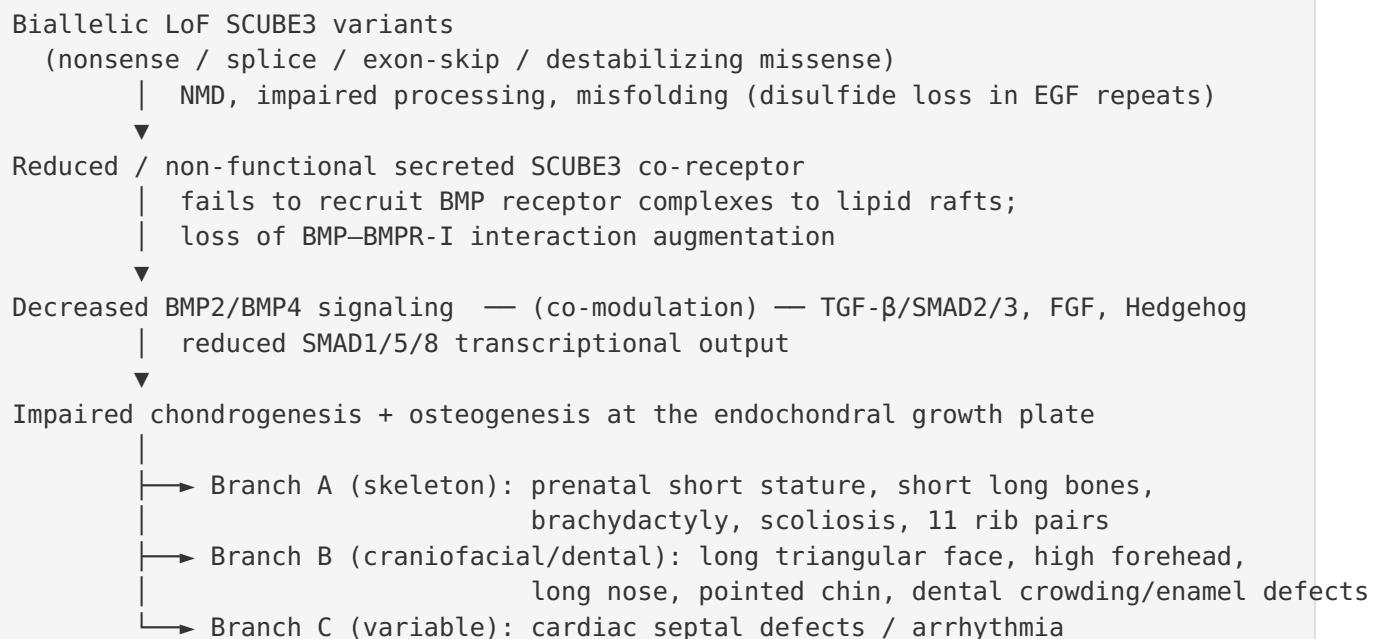
Model characteristics:

"Scube3^{-/-} mice showed craniofacial and dental defects, reduced body size, and defective endochondral bone growth due to impaired BMP-mediated chondrogenesis and osteogenesis, recapitulating the human disorder." — PMID: [33308444](#)

"Scube3^{N294K/N294K} mutants showed morphological abnormalities of the skeleton, alterations of parameters relevant for bone metabolism, changes in renal function, and hearing impairments." — PMID: [27815347](#)

Applications: growth-plate BMP signaling, skeletal/craniofacial/dental development, bone metabolism. **Limitations:** the knockout captures core skeletal/craniofacial phenotypes but human variant-specific hypomorphic effects (variable secretion/function) and the human cognitive-sparing may not be fully modeled by null alleles.

Mechanistic Model / Interpretation



Upstream vs downstream: the mutation → protein loss → BMP co-receptor failure steps are upstream and directly demonstrated in vitro and in the mouse; the tissue-specific clinical branches are downstream and reflect the developmental requirement for BMP signaling in each lineage. The consistency between the in-vitro co-receptor data, the AlphaFold3 structural modeling, and the faithful *Scube3*^{-/-} mouse gives this causal chain strong internal coherence.

Evidence Base

PMID	Title (abbrev.)	Role
33308444	<i>SCUBE3 loss-of-function causes a recognizable recessive developmental disorder due to defective BMP signaling</i>	Landmark/defining paper. Cohort of 18 individuals/9 families; establishes recessive inheritance, phenotype, BMP2/BMP4 co-receptor mechanism, in-vitro functional validation, and the recapitulating <i>Scube3</i> ^{-/-} mouse.
40331102	Yeter et al. 2025 case report	Expands phenotype (ocular/skeletal), reports ~20 patients worldwide, AlphaFold3 disulfide-disruption mechanism for p.Cys303Ser.
37237303	SCUBE3 review	Defines five-domain protein architecture, expression sites, multipathway co-receptor role, vertebrate conservation.
21441952	<i>SCUBE3 is an endogenous TGF-β receptor ligand...</i>	Documents CUB-domain TGF-β binding → SMAD2/3 phosphorylation—the TGF-β arm of SCUBE3 function.
39250278	<i>SCUBE3 promotes osteogenic differentiation... in hBMSCs</i>	In-vitro confirmation of pro-osteogenic BMP2/TGF-β-SMAD role; supports the growth/bone mechanism.
27815347	<i>Scube3</i> ^{N294K} ENU mouse	Independent missense mouse model with skeletal/bone-metabolism phenotype.
24084593	<i>Scube regulates synovial angiogenesis-related signaling</i>	Contextualizes SCUBE family antagonism/agonism across BMP/TGF-β/Hedgehog.
41001785	Korean SGA-SS exome study	Shows <i>SCUBE3</i> appears among genes with likely-pathogenic variants in short-stature diagnostic panels.

All primary mechanistic and clinical claims here are **human clinical** (cohort/case reports) reinforced by **model-organism** (mouse, zebrafish) and **in-vitro/computational** (functional assays, AlphaFold3) evidence.

Limitations and Knowledge Gaps

- **Small evidence base:** ~20 patients worldwide; phenotype frequencies derive from a single founding cohort (~15 evaluable individuals). Frequencies are precise-looking but statistically fragile.
 - **No epidemiology:** no prevalence/incidence estimate; no Orphanet ORPHA or MeSH term.
 - **No prognostic biomarkers** and no natural-history/longitudinal cohort data (e.g., adult final height distributions, scoliosis progression rates).
 - **Genotype–phenotype correlations** are not yet established (whether hypomorphic missense vs null alleles predict severity).
 - **GH therapy efficacy is untested rigorously**—used empirically with unproven benefit.
 - **TGF- β /FGF/Hedgehog contributions** to the human phenotype are inferred; only the BMP arm is directly demonstrated for the disease.
 - **Cognitive-sparing mechanism** (why SCUBE3 loss largely spares cognition despite CNS expression) is unexplained.
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Proposed Follow-up Experiments / Actions

1. **Establish an international patient registry** to capture natural history: final adult height, scoliosis progression, cardiac and dental outcomes, and QoL (SF-36/PROMIS).
2. **Genotype–phenotype study** correlating residual SCUBE3 secretion/function (per-variant in-vitro assays) with clinical severity to build a variant-effect map on NM_152753.4.
3. **Controlled evaluation of recombinant GH** (or, mechanistically, growth-plate-directed BMP-pathway modulation) in a small cohort, given the local BMP defect rationale.
4. **Conditional/tissue-specific *Scube3* mouse models** (chondrocyte-, neural-crest-, cardiac-specific) to dissect the phenotype branches and test BMP-augmentation rescue.
5. **Deep phenotyping of the cardiac and ocular spectrum** to refine "with or without cardiac anomalies" penetrance estimates.
6. **Structural/biochemical work** on how missense EGF-repeat disulfide disruptions impair secretion, to identify potentially rescuable (chaperone-amenable) alleles.
7. **Formal ClinGen/ACMG variant curation and gene-disease validity classification** to standardize diagnostic reporting.

Report compiled from 10 confirmed findings across 5 investigation iterations. Evidence sources: human clinical cohorts/case reports, mouse and zebrafish model organisms, in-vitro functional assays, and computational structural modeling.

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